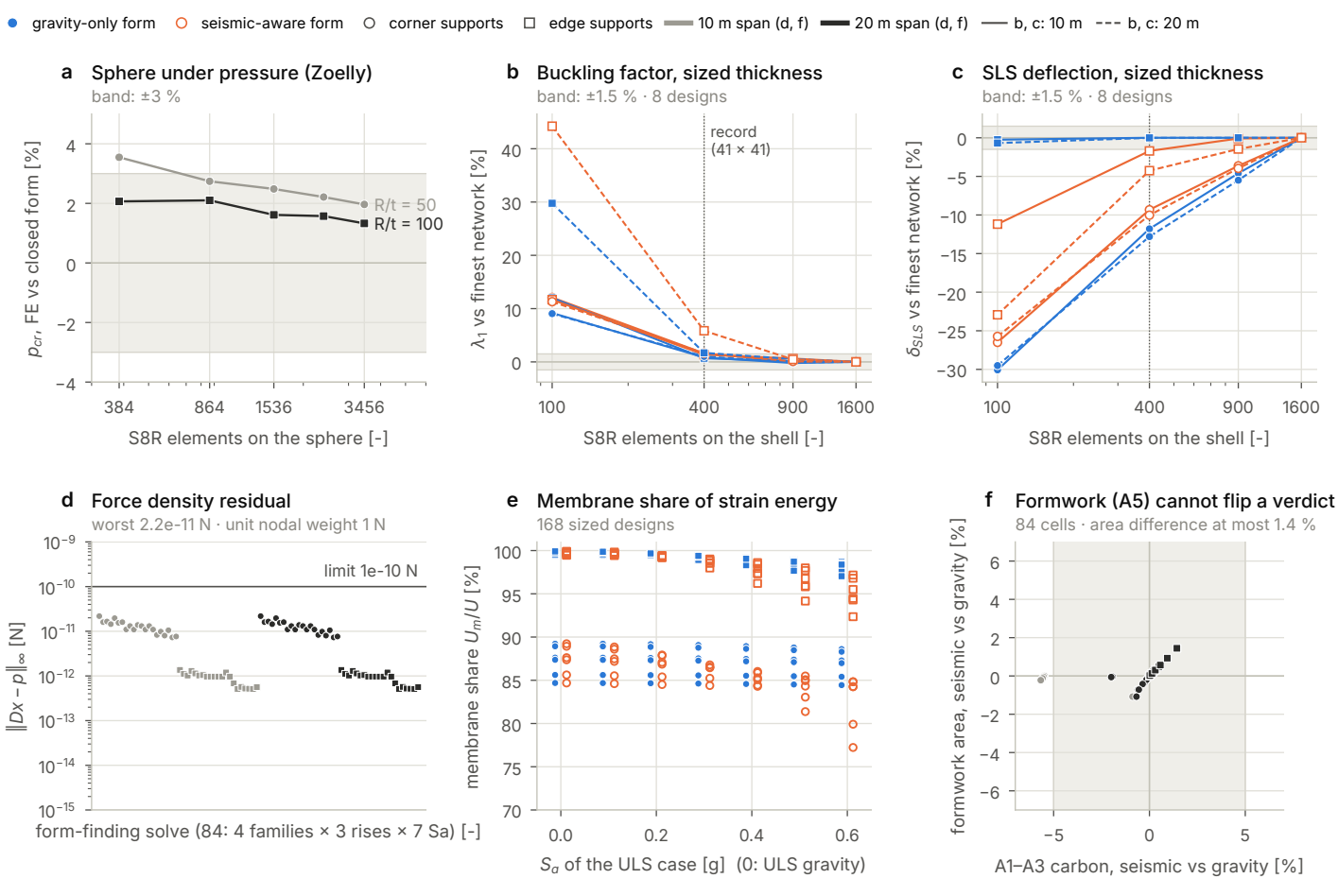

gravity-only form
seismic-aware form
corner supports
edge supports
10 m span (d, f)
20 m span (d, f)
b, c: 10 m
b, c: 20 m
a
Sphere under pressure (Zoelly)
band: ±3 %
p_{cr}, FE vs closed form [%]
R/t = 50
R/t = 100
S8R elements on the sphere [-]
b
Buckling factor, sized thickness
band: ±1.5 % · 8 designs
λ_1 vs finest network [%]
record
(41 × 41)
S8R elements on the shell [-]
c
SLS deflection, sized thickness
band: ±1.5 % · 8 designs
δ_{SLS} vs finest network [%]
S8R elements on the shell [-]
d
Force density residual
worst 2.2e-11 N · unit nodal weight 1 N
limit 1e-10 N
$\|Dx - p\|_\infty$ [N]
form-finding solve (84: 4 families × 3 rises × 7 Sa) [-]
e
Membrane share of strain energy
168 sized designs
membrane share U_m/U [%]
S_a of the ULS case [g] (0: ULS gravity)
f
Formwork (A5) cannot flip a verdict
84 cells · area difference at most 1.4 %
formwork area, seismic vs gravity [%]
A1–A3 carbon, seismic vs gravity [%]